(5) $L_x = \frac{d^2}{dt^2} + 1.$

We have to solve the eqn:

$$L_x G(t, \varepsilon) = \delta(t - \varepsilon)$$

(a) $t \neq \varepsilon$: $\frac{d^2 G}{dt^2} + G = 0.$

$$\Rightarrow \boxed{G'' + G = 0}$$

General soln:

$$G(t, \varepsilon) = \begin{cases} A_1 \cos t + A_2 \sin t & ; t < \varepsilon \\ B_1 \cos t + B_2 \sin t & t > \varepsilon \end{cases}$$

(i) Continuity @ $t = \varepsilon$:

$$G(t, \varepsilon^+)\Big|_{t=\varepsilon^+} = G(t, \varepsilon)\Big|_{t=\varepsilon^-}$$

$$\Rightarrow A_1 \cos\varepsilon + A_2 \sin\varepsilon = B_1 \cos\varepsilon + B_2 \sin\varepsilon \quad \text{—— (1)}$$

(ii) Boundary conditions:

$y(t=0) = 0 \Rightarrow G(t=0) = 0.$

$\Rightarrow \boxed{A_1 = 0}$

$y'(t=0) = 0 \Rightarrow G'(t=0) = 0.$

$\Rightarrow -A_1 \sin t + A_2 \cos t \big|_{t=0} = 0.$

$\Rightarrow \boxed{A_2 = 0}$

$$\therefore G(t, \varepsilon) = \begin{cases} 0 & ; t < \varepsilon \\ B_1 \cos t + B_2 \sin t & ; t > \varepsilon \end{cases}$$

From (1) : $\boxed{B_1 \cos\varepsilon = -B_2 \sin\varepsilon}$ —— (m)

(iii) Derivative jump @ $t = \varepsilon$:

$$\int_{\varepsilon-\eta}^{\varepsilon+\eta} G'' \, dt + \cancel{\int_{\varepsilon-\eta}^{\varepsilon+\eta} G \, dt}^{\to 0} = \int_{\varepsilon-\eta}^{\varepsilon+\eta} \delta(t-\varepsilon) \, dt$$

$$\Rightarrow \quad G' \Big|_{\varepsilon-\eta}^{\varepsilon+\eta} = 1 \qquad \left[\eta \to 0 \text{ so integral becomes } 0. \quad \int_{t'-\varepsilon}^{t'+\varepsilon} \delta(t-t') \, dt = 1 \right]$$

$$\Rightarrow \quad G'(\varepsilon+\eta) - G'(\varepsilon-\eta) = 1.$$

$$\therefore \quad G'(t,\varepsilon) = \begin{cases} 0 & ; \quad t < \varepsilon \\ -B_1 \sin t + B_2 \cos t & ; \quad t > \varepsilon \end{cases}$$

$$\therefore \quad G'(\varepsilon+\eta) - G'(\varepsilon-\eta)$$

$$\Rightarrow \quad \boxed{-B_1 \sin \varepsilon + B_2 \cos \varepsilon = 1} \quad \text{— (iii)}$$

from (ii) : $B_1 = -B_2 \dfrac{\sin \varepsilon}{\cos \varepsilon}$

Substituting in (iii) :

$$B_2 \sin^2 \varepsilon + B_2 \cos^2 \varepsilon = \cos \varepsilon \Rightarrow \boxed{B_2 = \cos \varepsilon} \quad \text{— (iv)}$$

$$\therefore \quad \boxed{B_1 = \cancel{-B} \; -\sin \varepsilon}$$

$$\therefore \quad G(t > \varepsilon) = -\sin \varepsilon \cos t + \cos \varepsilon \sin t$$

$$= \sin(t-\varepsilon) = \sin(t-t') \qquad [\varepsilon \to t']$$

$$\therefore \quad G(t,t') = \begin{cases} 0 & ; \quad t < t' \\ \sin(t-t') & ; \quad t > t' \end{cases}$$

$$\boxed{G(t,t') = \theta(t-t') \sin(t-t')} \quad \text{(Ans)}$$

(4) We have to solve for:

$$\frac{d^2 G}{dx^2}(x,x') = \delta(x-x') \quad ; \quad G(x,x') \to \text{Green's } f^n.$$

$$f(x) = \int_0^1 G(x,x') \, g(x') \, dx'$$

@ $x \neq x'$: $G'' = 0$

General sol^n: $G(x,x') = \begin{cases} A_1 x + A_2 & ; \; 0 \leq x < x' \\ B_1 x + B_2 & ; \; 1 \geq x > x' \end{cases}$

(i) ~~Conti~~ Boundary conditions :

(14)

$f(0) = 0 \Rightarrow G(x=0) = 0$

$\Rightarrow \boxed{A_2 = 0}$ — (i)

$f(1) = 0 \Rightarrow G(x=1) = 0$

$\Rightarrow B_2 + B_1 = 0$

$\Rightarrow \boxed{B_2 = -B_1}$ — (ii)

$$G(x,x') = \begin{cases} A_1 x & ; \quad x < x' \\ B_1 (x-1) & ; \quad x > x' \end{cases}$$

(ii) Continuity @ $x = x'$:

$G(x'+) = G(x'-)$

$\Rightarrow B_1(x'-1) = A_1 x'$

$\Rightarrow \boxed{x'(B_1 - A_1) = B_1}$ — (iii)

(iii) Derivative jump @ $x = x'$:

$$\int_{x'-\varepsilon}^{x'+\varepsilon} \frac{d^2G}{dx^2}\, dx = \int_{x'-\varepsilon}^{x'+\varepsilon} \delta(x-x')\, dx$$

$\Rightarrow G'(x'+\varepsilon) - G'(x'-\varepsilon) = 1$

$\Rightarrow B_1 - A_1 = 1 \Rightarrow \boxed{B_1 = 1 + A_1}$ — (iv)

Substituting (iv) in (iii):

$\boxed{B_1 = x'}$; $\boxed{A_1 = x'-1}$.

$$\therefore G(x,x') = \begin{cases} x(x'-1) & ; \quad x < x' \\ x'(x-1) & ; \quad x > x' \end{cases}$$

$$\therefore f(x) = \int_0^1 G(x,x')\, g(x')\, dx'$$

$$= \int_0^{x} x'(x-1)\, g(x')\, dx' + \int_x^1 x(x'-1)\, g(x')\, dx'$$

$$\boxed{f(x) = (x-1)\int_0^x x'\, g(x')\, dx' + x\int_x^1 (x'-1)\, g(x')\, dx'}$$ (Ans)

③ $L\psi(x) = \lambda\psi(x)$

λ — eigenvalue ; $\psi(x)$ — eigenvector.

~~$\omega(x) =$~~ $L = x\frac{d^2}{dx^2} + (1-x)\frac{d}{dx}$.

$p_0 = x$; $p_1 = (1-x)$.

For L to be self-adjoint ;

$p_0' = p_1$

$p_0' = 1 \neq p_1$.

Hence L is not self-adjoint.

$$\omega(x) = p_0^{-1} \exp\left[+\int \frac{p_1}{p_0}\,dx\right]$$

$$= \frac{1}{x}\exp\left[+\int \frac{1-x}{x}\,dx\right].$$

$$= \frac{1}{x}\exp\left[0 - x + \ln x\right]$$

$$= \frac{1}{x}\underbrace{e^{-x}}_{\bar{p}_0}\cdot e^{+\ln x} = \boxed{e^{-x} = \omega(x)} \quad \text{(Ans)}$$

$$\therefore \omega(x)L = \underbrace{e^{-x}x}_{\bar{p}_0}\frac{d^2}{dx^2} + \underbrace{e^{-x}(1-x)}_{\bar{p}_1}\frac{d}{dx}.$$

$$= \frac{d}{dx}\left[xe^{-x}\frac{d}{dx}\right] \quad ; [\bar{p}_0' = \bar{p}_1]$$

$$\int_0^\infty v^*(\omega L)u\,dx. \quad ; \quad u, v \text{ — eigenf}^{n}\text{s of } \omega L$$

$$= \int_0^\infty \underset{①}{v^*}\,\overbrace{\frac{d}{dx}\left(xe^{-x}\frac{du}{dx}\right)}^{⑪}\,dx.$$

$$= v^* xe^{-x}\frac{du}{dx}\Big|_0^\infty - \int \overbrace{\frac{dv^*}{dx}xe^{-x}}^{①}\,\overbrace{\frac{du}{dx}}^{⑪}\,dx.$$

$$= [\qquad\qquad] - \frac{dv^*}{dx}xe^{-x}u\Big|_0^\infty + \int_0^\infty \frac{d^2}{dx}\left(\frac{dv^*}{dx}xe^{-x}\right)u\,dx$$

$$= \underbrace{\left[v^* xe^{-x}\frac{du}{dx} - \frac{dv^*}{dx}xe^{-x}u\right]_0^\infty}_{\text{Boundary-term.}} + \underbrace{\int\left\{\frac{d}{dx}\left(xe^{-x}\frac{d}{dx}\right)v\right\}^* u\,dx}_{\int_0^\infty (Lv)^* u\,dx.}$$

$u(x\to\infty) = 0$ ~~$u(x=0)$~~ [Dirichlet BC]

$v(x\to\infty) = 0$

$\therefore$ @ $x\to\infty$: $v^* xe^{-x}\frac{du}{dx} = \frac{dv^*}{dx}xe^{-x}u = 0$ [$u\to 0$; $e^{-x}\to 0$]

@ $x = 0$: $v^* xe^{-x}\frac{du}{dx} - \frac{dv^*}{dx}xe^{-x}u = 0$ [$x=0$]

Hence the Boundary term vanishes.

$$\boxed{\int_0^\infty x e^{-x} u_n u_m = \delta_{mn}}$$ — New definat^n of orthogonality.